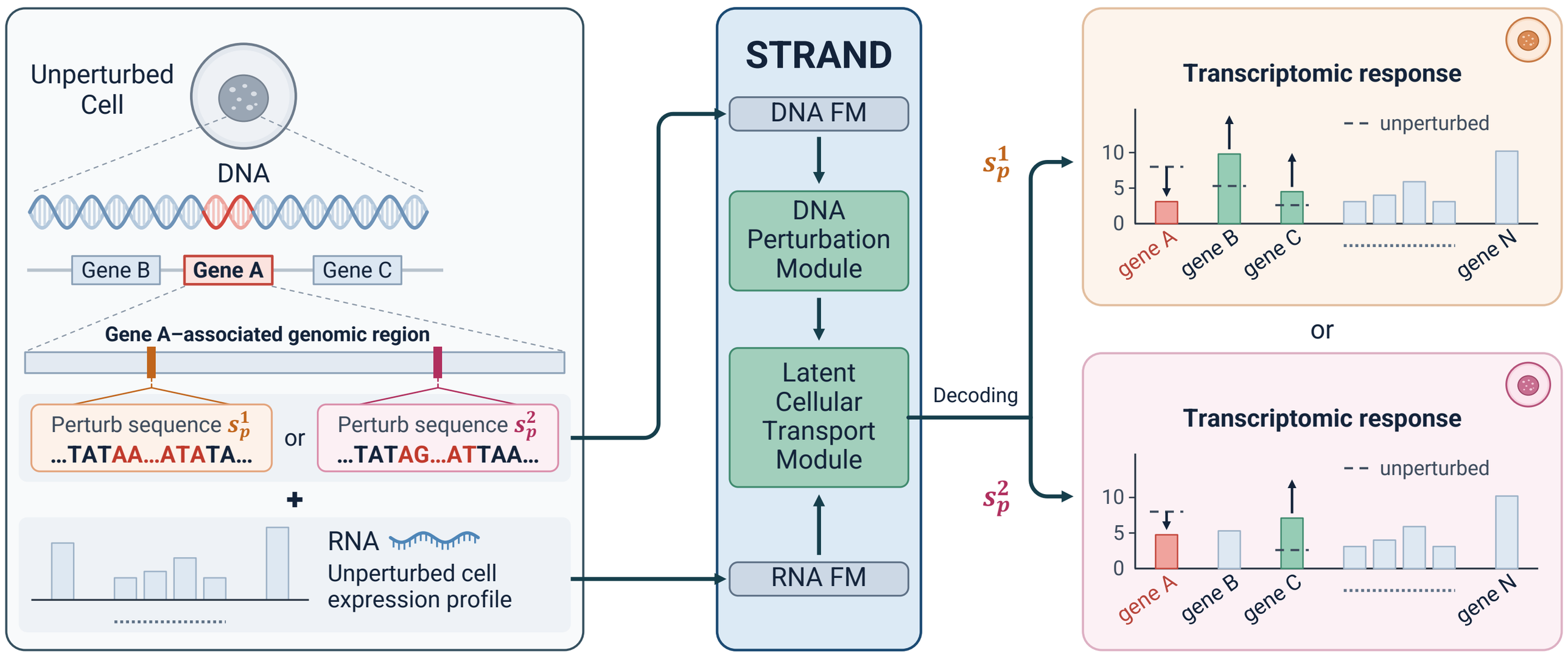

Unperturbed Cell
DNA
Gene B
Gene A
Gene C
Gene A–associated genomic region
Perturb sequence s_p^1
...TATAA...ATATA...
or
Perturb sequence s_p^2
...TATAG...ATTAA...
+
RNA
Unperturbed cell expression profile
STRAND
DNA FM
DNA Perturbation Module
Latent Cellular Transport Module
RNA FM
Decoding
s_p^1
s_p^2
Transcriptomic response
unperturbed
10
5
0
gene A
gene B
gene C
gene N
or
Transcriptomic response
unperturbed
10
5
0
gene A
gene B
gene C
gene N